

BLE Series

Data Sheet

USE_8762_MINI
-BT5.0 BLE Module

Version

VERSION	REVISION DATE	REVISOR	REVIEWER	REVISED CONTENTS
1.0	20/08 2024	MB	AK	Mini BLE Module
2.0	25/09/2024	DJ	AK	Mini BLE Module New Design
2.1	27/09/2024	AB	AK	Additional Parameters
2.2	28/02/2025	DJ	AK	Recommended Connections
2.3	15/05/2025	AB	AK	GPIO pins descriptions
2.4	04/08/2025	DJ	AD	Module placement strategy
2.5	10/09/2025	DJ	AK	Module Pad Dimensions
2.6	20/01/2026	AD	DJ	Added Product Dimension

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Features:

General

- 3.3 V Operational power supply
- Supports Bluetooth 5.0 core specification
- Supports 2Mbps LE
- LE advertising Extensions
- LE Long Range
- Additional Adv channel
- Channel Selection #2
- High Duty Cycle Non-Connectable Adv
- Integrated MCU to execute Bluetooth protocol stack
- Supports multiple-level Low Energy states
- Supports LE L2CAP Connection Oriented Channel Support
- Supports LE low-duty directed advertising
- Supports LE data length extension feature
- Supports OTA (Over-the-Air) programming mechanism for firmware upgrade
- Supports GAP, ATT/GATT, SMP, L2CAP
- Generic Applications for GAP Central, Peripheral, Observer and Broadcaster Roles
- Ultra-low power consumption with intelligent PMU

MCU Platform

- ARM Cortex-M4 CPU with floating-point unit (Maximum 40MHz)
- Serial flash controller (One and Quad-bits mode) with 16kB 4-way cache
- Total 160kB SRAM
- 4Kbits eFUSE for manufacturer use
- Supports AES128/192/256 encrypt/decrypt engine
- Embedded 4 Mbits Flash

Bluetooth Transceiver

- Tx Maximum Output Power: 8dBm
- RX sensitivity: -98.5dBm BLE
- Fast AGC control to improve receiving dynamic range
- Supports Bluetooth Low Energy PHY

Peripheral Interfaces

- 12 flexible general purpose IOs (maximum).
- Hardware Keyscan
- Embedded IR transceiver
- Supports generic 4-wire SPI master/slave
- Supports 6-channel low-power comparators
- 400kbps, 12-bit, maximum 6-channel AUXADC
- USE_8762_MINI only supports 2-channel
- Timers x 8
- I2C x 2

- PWM x 8
- UART x 2
- Supports external 40MHz XTAL without capacitors (in limited conditions)

General Description:

The USE_8762_MINI is an ultra-low-power system-on-chip solution for Bluetooth 5.0 low energy applications that combines the excellent performance of a leading RF transceiver with a low-power ARM Cortex-M4F along with rich, powerful supporting features and peripherals like Power Management Unit (PMU), ADC, and smart I/O distribution controller.

For more information reach out to us at info@unisemindia.com.

Pin Definition:

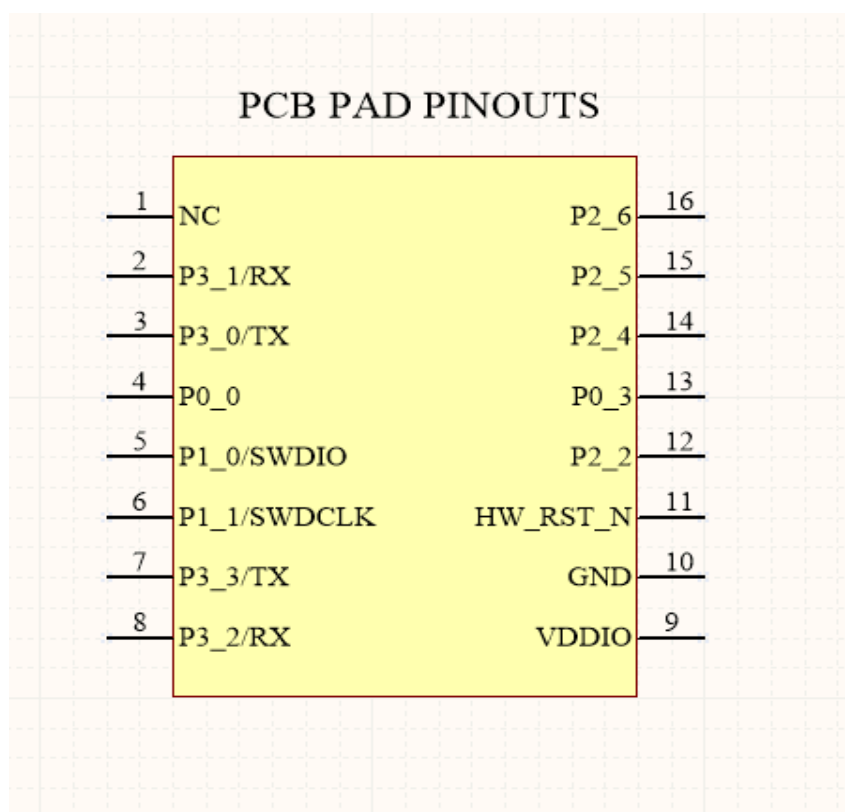


Fig 1 : Module Pinouts

On the top-left side soldering pad marked as '1' and bottom left side marked as '8', on the bottom-right side, the soldering pad marked as '9' and top-right side marked as '16'.

For more information refer to Module Diagram (**Fig 6**).

Pin Descriptions:

Pin No.	Pin Name	Type	Function Description
1	NC	-	No Connection
2	P3_1	IO/UART_RX	General purpose IO;Pin Multiplexer.8mA driving capability. Includes wakeup function. Has internal strong/weak pull-up and pull-down.HCI_UART_RX (default).
3	P3_0	IO/UART_TX	General purpose IO;Pin Multiplexer.8mA driving capability. Includes wakeup function. Has internal strong/weak pull-up and pull-down. HCI_UART_TX (default).
4	P0_0	IO	General purpose IO;Pin Multiplexer.8mA driving capability. Includes wakeup function. Has internal strong/weak pull-up and pull-down. Enable the DLPS when grounded.
5	P1_0	SWDIO	General purpose IO;Pin Multiplexer.8mA driving capability. Includes wakeup function. Has internal strong/weak pull-up and pull-down. SWDIO (default).
6	P1_1	SWDCLK	General purpose IO;Pin Multiplexer.8mA driving capability. Includes wakeup function. Has internal strong/weak pull-up and pull-down. SWCLK (default).
7	P3_3	IO/UART_TX	General purpose IO;Pin Multiplexer.8mA driving capability. Includes wakeup function. Has internal strong/weak pull-up and pull-down. AT_CMD_UART_TX.
8	P3_2	IO/UART_RX	General purpose IO;Pin Multiplexer.8mA driving capability. Includes wakeup function. Has internal strong/weak pull-up and pull-down. AT_CMD_UART_RX.
9	VDDIO	PWR	Supply 1.8V~3.3V power.
10	GND	GND	E-PAD is a ground pin in the QFN package.
11	HW_RST_IN	I	Hardware reset pin; active low
12	P2_2	IO	General purpose IO; Pin Multiplexer.8mA driving capability. Includes wake-up function. Has internal strong/weak pull-up and pull-down.
13	P0_3	IO	LOG_UART TX. Power on trap: pull-up for normal operation. Pull-down to bypass executing program code in flash. (PAD internal pull-up by default)

14	P2_4	IO	General purpose IO;Pin Multiplexer.8mA driving capability. Includes wakeup function. Has internal strong/weak pull-up and pull-down.
15	P2_5	IO	General purpose IO;Pin Multiplexer.8mA driving capability. Includes wakeup function. Has internal strong/weak pull-up and pull-down.
16	P2_6	IO	General purpose IO;Pin Multiplexer.8mA driving capability. Includes wakeup function. Has internal strong/weak pull-up and pull-down.

Power Supply DC Characteristics:

Symbol	Parameter	Min	Typ	Max	Units
VDDIO	Single Power Source for Whole Chip	2.4	3	3.6	V
POR (Power on Reset)	Rising time from 0V to VBAT(1.8V) recommended to be less than 540μs.	-	1.72	1.78	V
BOR (Brown out Reset)	Falling time from VBAT(1.8V) to V _{BOR} recommended to be more than 30μs.	1.61	1.67	-	V

Digital IO Pin DC Characteristics:

Parameter	Condition	Min	Typ	Max	Units
Input High Voltage	VDDIO = 3.3V	2	3.3	3.6	V
Input Low Voltage	VDDIO = 3.3V	-	0	0.9	V
Output High Voltage	VDDIO = 3.3V	2.97	-	3.3	V
Output Low Voltage	VDDIO = 3.3V	0	-	0.33	V
Input High Voltage	VDDIO = 2.8V	1.8	2.8	3.1	V
Input Low Voltage	VDDIO = 2.8V	-	0	0.8	V
Output High Voltage	VDDIO = 2.8V	2.5	-	-	V
Output Low Voltage	VDDIO = 2.8V	0	-	0.28	V

Pull High and Pull Low Resistor	VDDIO = 3.3V Strong pull/weak pull	-	10/100	-	kΩ
	VDDIO = 1.8V Strong pull/weak pull	-	20/200	-	kΩ
	VDDIO = 3.3V Strong pull/weak pull (P2_2 $\downarrow$ P2_7, P5_0)	-	5/50	-	kΩ
	VDDIO = 1.8V Strong pull/weak pull (P2_2 $\downarrow$ P2_7, P5_0)	-	2.5/25	-	kΩ
Input High Current	PAD configured as input mode	-	-	0.1	μA
Input Low Current	PAD configured as input mode	-	-	0.1	μA

Power Modes:

The USE_8762_MINI defines three PMU power states for various conditions.

Active Mode: All clock and power is turned on. All functions operate in this mode.

Deep LPS Mode: High-speed clock and core domain power is turned off. The CPU stops running. Data can be retained in retention SRAM.

Power Down Mode: Except in an ‘always-on’ power domain, all clock sources and power are turned off. Power down mode can only be woken by GPIO pins.

Current Consumption on Low Power Mode (at 3V):

Power Mode	Always on Registers	32kHz RCOSC	Retention SRAM	CPU	Wakeup Method	Current Consumption (Typ.)
Power down	ON	OFF	OFF	OFF	Wakeup by GPIO	0.7μA
Deep LPS	ON	ON	Retention	OFF	Wakeup by GPIO, timer	3.81μA (with 160 KB SRAM in retention state)

Current Consumption on Active Mode (at 3V):

Power Mode	Always on Registers
Active RX mode	6.15 mA
Active TX mode (TX power: 0 dBm)	2.34mA
Active TX mode (TX power: 4 dBm)	2.39mA
Active TX mode (TX power: 7.5 dBm)	2.37mA

Temperature Limit Ratings:

Parameter	Min	Max	Units
Storage Temperature	-40	+105	°C
Ambient Operating Temperature	-40	+105	°C

ESD Characteristics:

Parameter	Condition	Min	Typ	Max	Units
HBM	All pins, test method: JESD22	-	-	+/- 4	kV

General Radio Characteristics:

Parameter	Condition	Min	Typ	Max	Units
Frequency Range	-	2402	-	2480	MHz

RX Performance:

Parameter	Condition	Min	Typ	Max	Units
Sensitivity	PER $\leq$ 30.8%	-	-98.5	-	dBm
Maximum Input Level	PER $\leq$ 30.8%	-	-1	-	dBm
C/I	C/I _{co-channel}	21	-	-	dBm
	C/I _{+1MHz}	15	-	-	dBm
	C/I _{-1MHz}	15	-	-	dBm
	C/I _{+2MHz}	-17	-	-	dBm
	C/I _{-2MHz}	-15	-	-	dBm
	C/I _{+3MHz}	-27	-	-	dBm
	C/I _{image}	-9	-	-	dBm
	C/I _{image+1MHz}	-15	-	-	dBm
	C/I _{image-1MHz}	-15	-	-	dBm
Blocking Power	30 $\cap$ 2000MHz, Wanted signal level = -67dBm	-30	-	-	dBm
	2003 $\cap$ 2399MHz, Wanted signal level = -67dBm	-35	-	-	dBm
	2484 $\cap$ 2997MHz, Wanted signal level = -67dBm	-35	-	-	dBm
	3000MHz $\cap$ 12.75GHz, Wanted signal level = -67dBm	-30	-	-	dBm
Max PER Report Integrity	Wanted signal: -30dBm	-	50	-	%
Max Intermodulation Level	Wanted signal (f ₀): -64dBm Worst intermodulation level @ 2f ₁ -f ₂ =f ₀ , f ₁ -f ₂ =n MHz, n=3, 4, 5...	-50	-	-	dBm

TX Performance:

Parameter	Condition	Min	Typ	Max	Units
Maximum Output Power	-	-	-	8	dBm
Adjacent Channel Power Ratio	+2MHz	-	-	-20	dBm
	-2MHz	-	-	-20	dBm
	>= +3MHz	-	-	-30	dBm
	<= -3MHz	-	-	-30	dBm
Modulation Characteristics	$\Delta f_{1_{avg}}$	-	250	-	kHz
	$\Delta f_{2_{max}}$	185	-	-	kHz
	$\Delta f_{2_{max}}$ pass rate	-	100	-	%
	$\Delta f_{2_{avg}} / \Delta f_{1_{avg}}$	-	0.88	-	-
Carrier Frequency Offset and Drift	Average F_n	-	12.5	-	kHz
	Drift rate	-	10	-	kHz/50 μ s
	Avg. drift	-	10	-	kHz/50 μ s
	Max. drift	-	10	-	kHz/50 μ s
Output Power of Second Harmonic	-	-	-50 ^{Note}	-	dBm
Output Power of Third Harmonic	-	-	-50 ^{Note}	-	dBm

Antenna Specification Details:

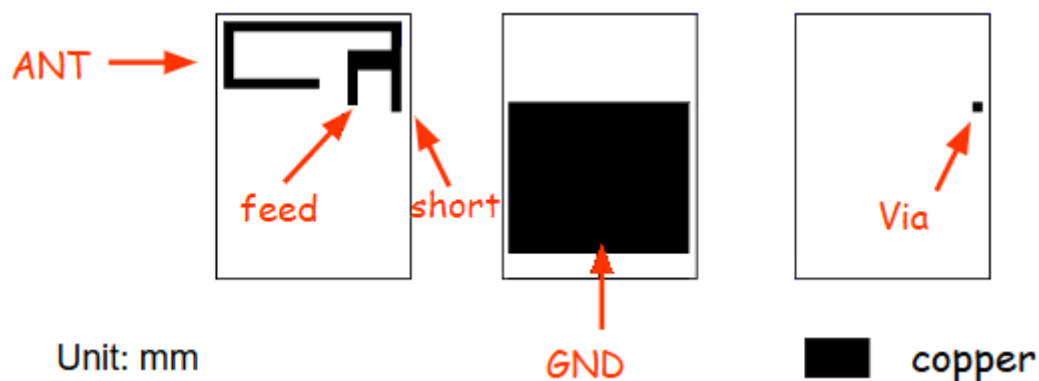


Fig 2 : Sample Antenna Structure & Footprint

Antenna Return Loss:

It describes the amount of power that is reflected back from the antenna due to impedance mismatches. The lower the return loss, the better the antenna is at efficiently radiating the signal, with minimal reflected power.

- Return Loss - 10dB-BW: 100 MHz

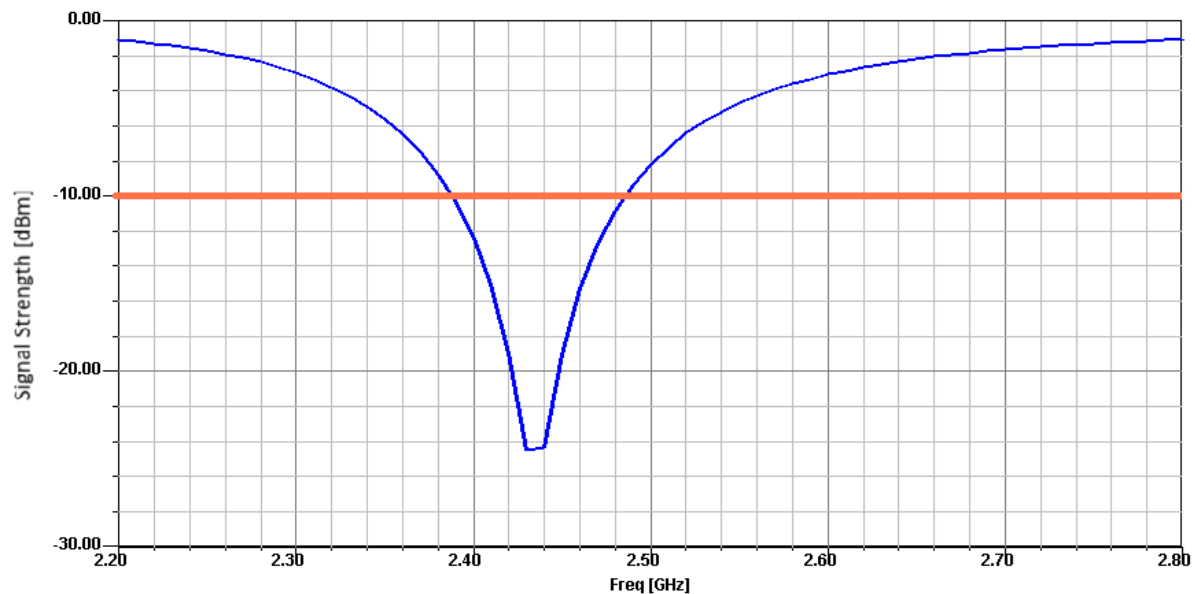


Fig 3

Antenna Radiation Patterns:

It describes how an antenna radiates or receives electromagnetic waves in space. It shows the distribution of radiated power as a function of direction in 3D space or a 2D plane. Radiation patterns help visualize how effectively an antenna transmits or receives signals in different directions.

- Radiation Patterns
 - Freq. : 2402 MHz Peak gain: 1.18 dBi
 - Freq. : 2440 MHz Peak gain: 1.30 dBi
 - Freq. : 2480 MHz Peak gain: 1.22 dBi

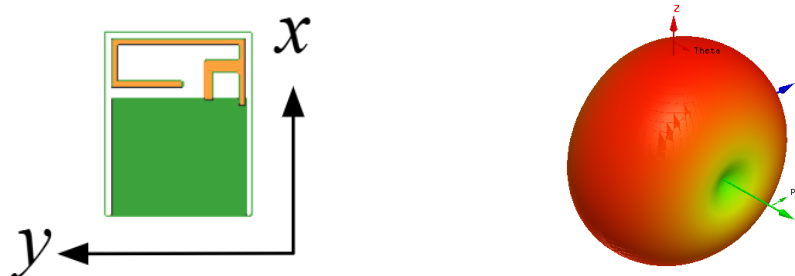


Fig 4

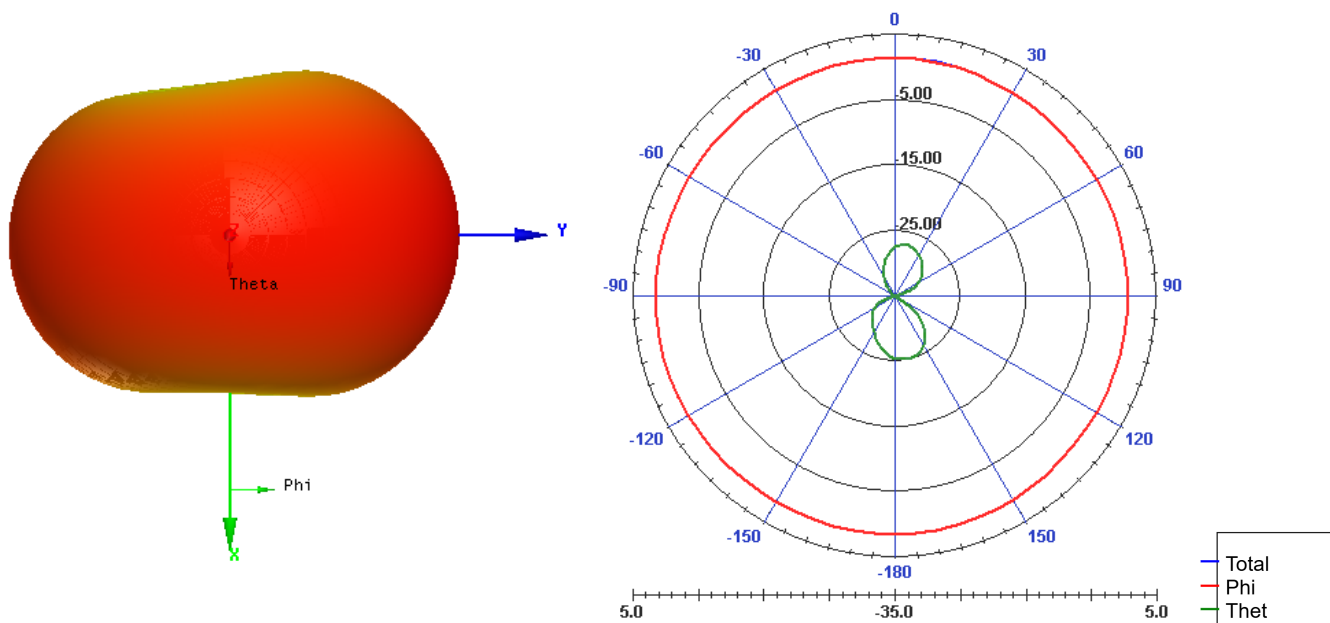


Fig 5(a) : yz-plane

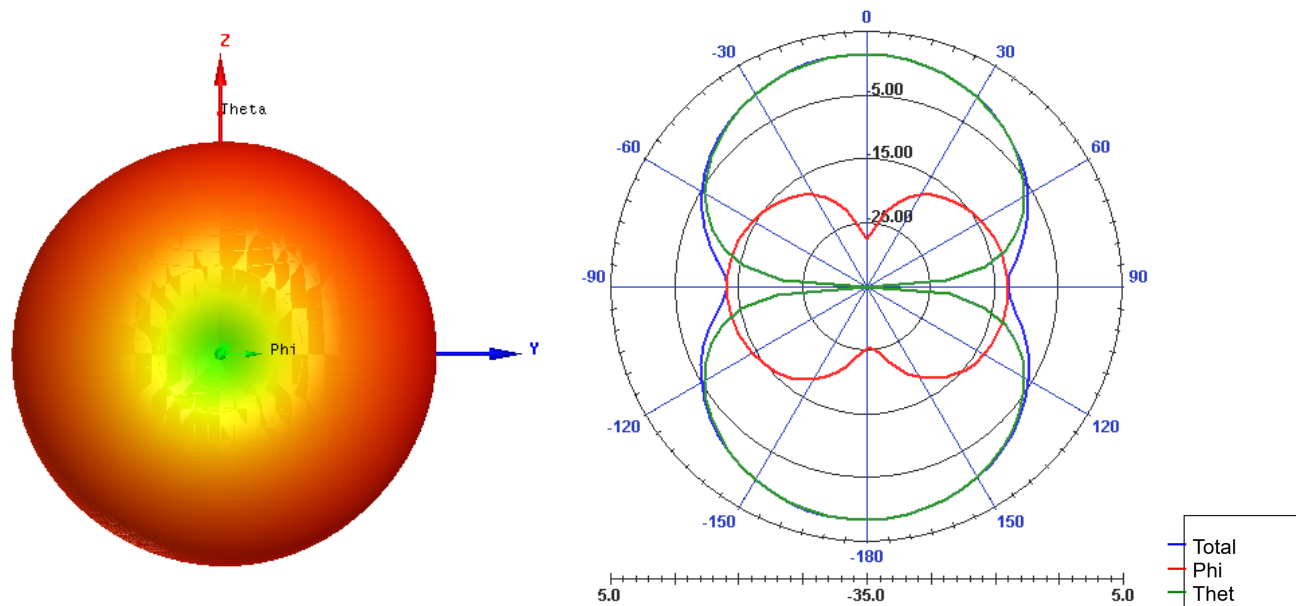


Fig 5(b) : xz-plane

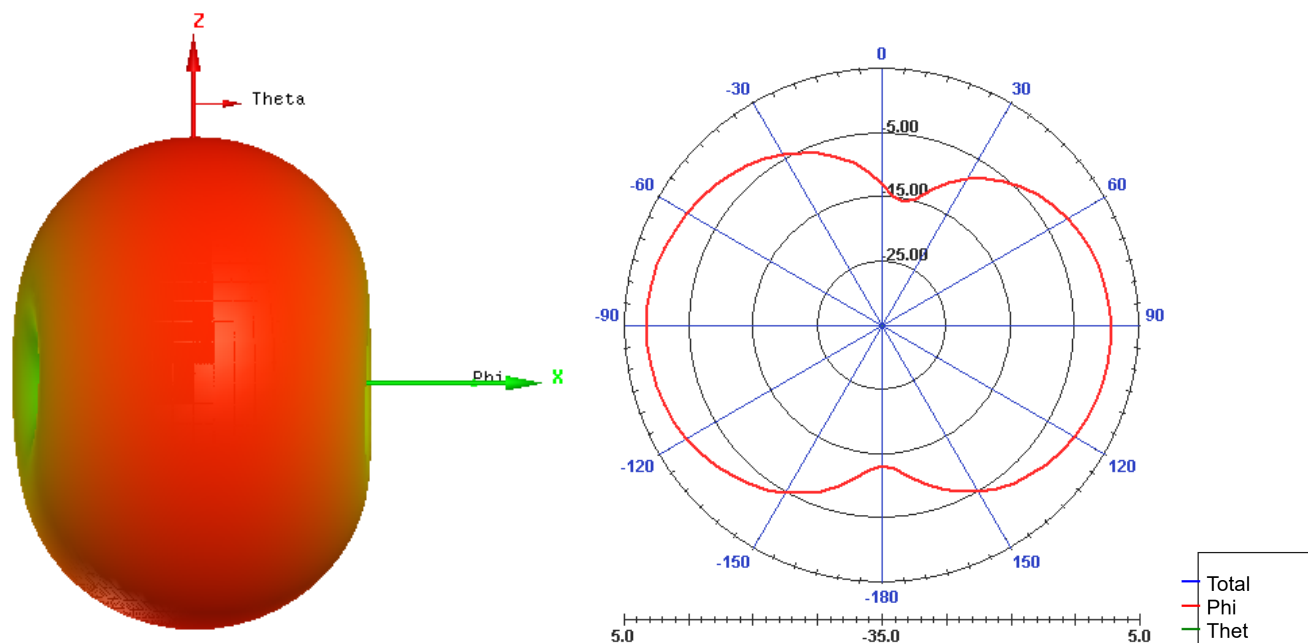


Fig 5(c) : xy-plane

Absolute Maximum Ratings:

Parameter	Min	Max	Units
Supply Voltage	2.4	3.6	V
Voltage on all other pins	0.33	3.3	V
Storage Temperature	-40	+105	°C
Output RF Power		8	dBm



- Stresses beyond the Absolute Maximum Ratings may cause permanent damage to the device
- These are stress ratings only, not the recommended operating ratings. Please refer Recommended Operating Condition section for safe functional operation range
- Exposure to Absolute Maximum Rating conditions for extended periods may affect device reliability
- All voltage values are with respect to ground

Recommended Operating Conditions:

Parameter	Min	Max	Units
Supply Voltage	2.4	3.3	V
Ambient Operating Temperature	-40	+105	°C

Applications:

- TV Remote Controller
- LE HID
- Beacon
- Home Automation
- Key Fob
- Toy

Module Diagram:

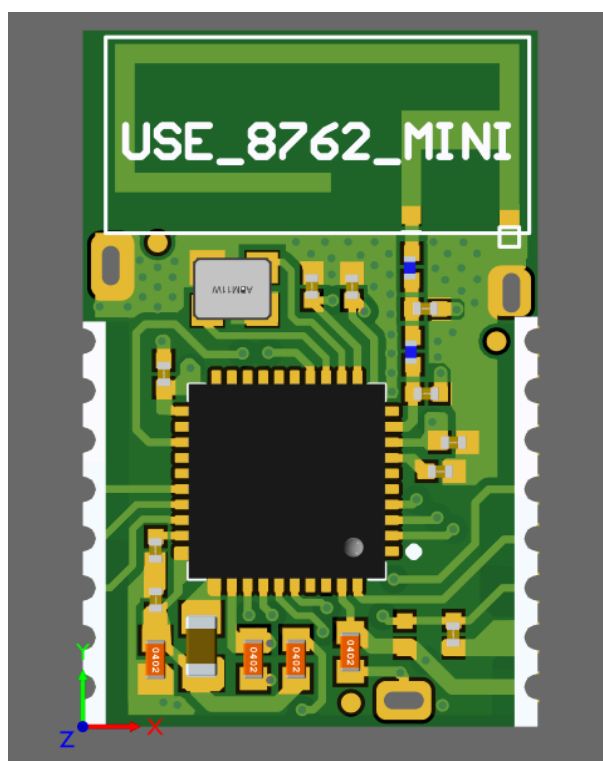


Fig 6 : Module without shield

Module Dimension Drawing:

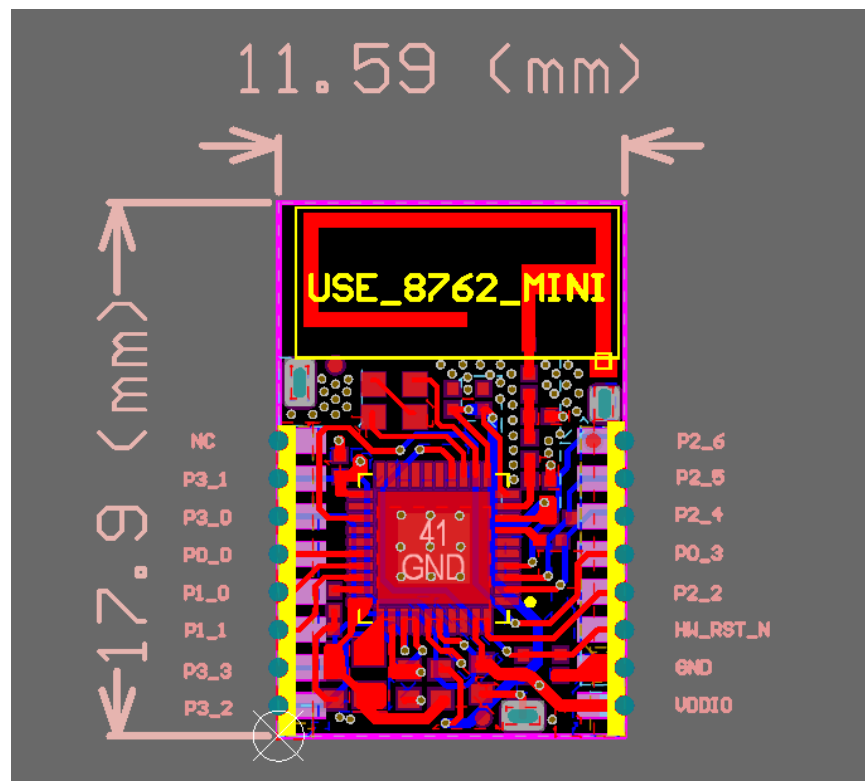


Fig 7(a) : Module Pinouts & Dimension

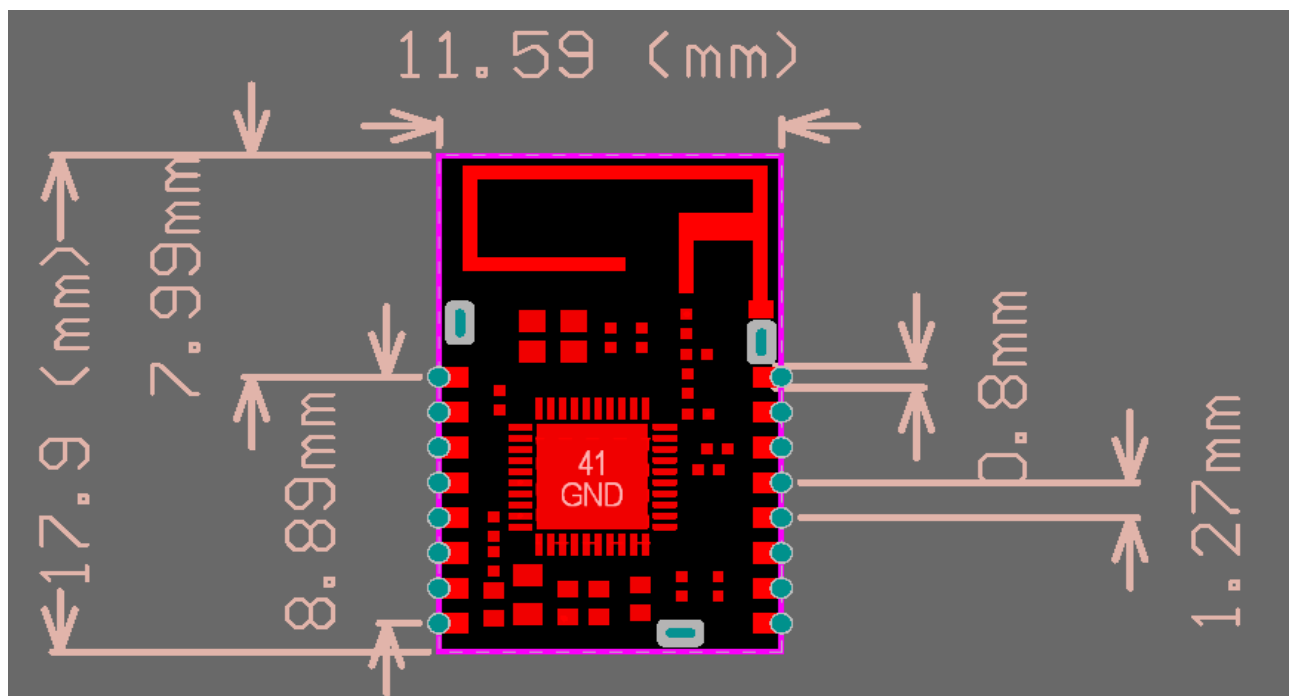


Fig 7(b) : Module Pad Dimension

Recommended Footprint Dimension:

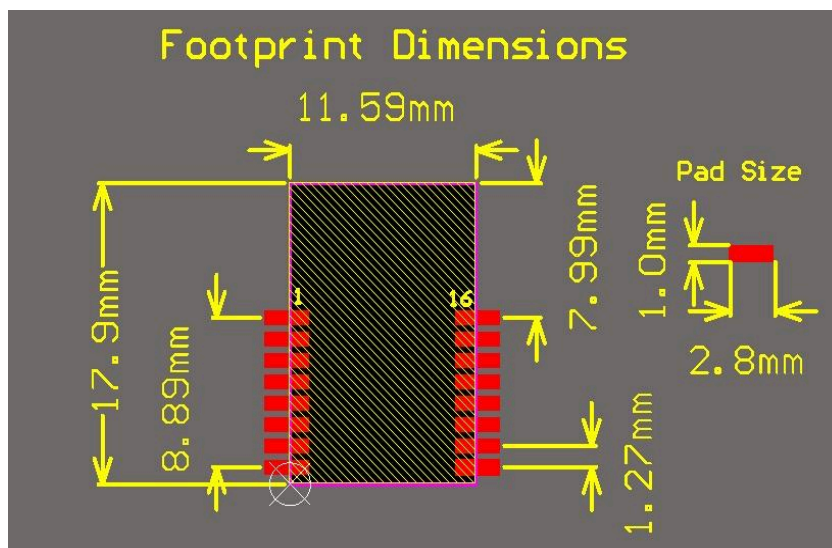


Fig 8 : Module Footprint Dimension

Product Dimension:

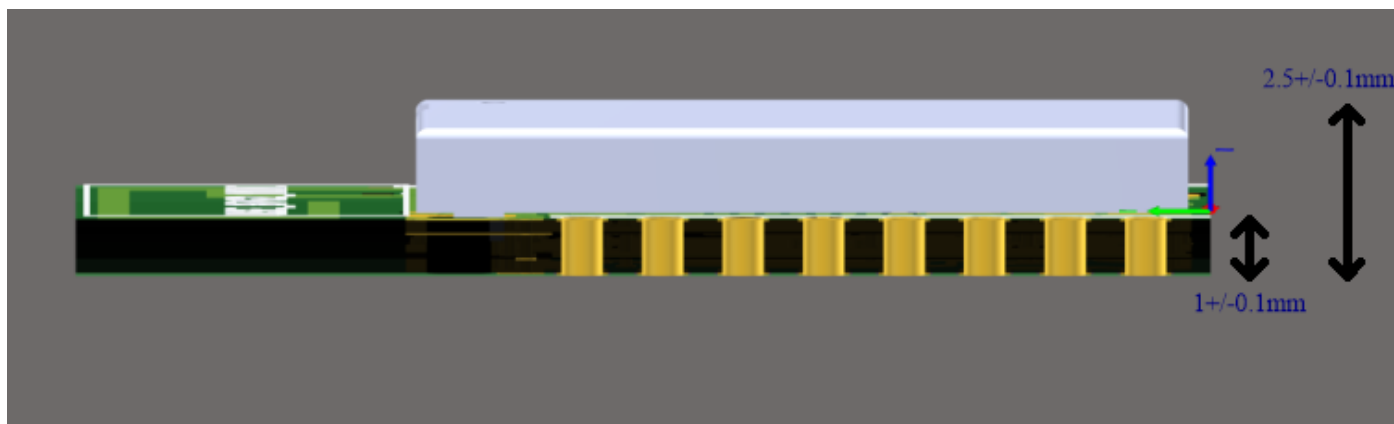


Fig 9 : Product Dimension

PCB Thickness : 1 ± 0.1 mm

Product Height (Including shield) : 2.5 ± 0.1 mm

Recommended Footprint Clearance:

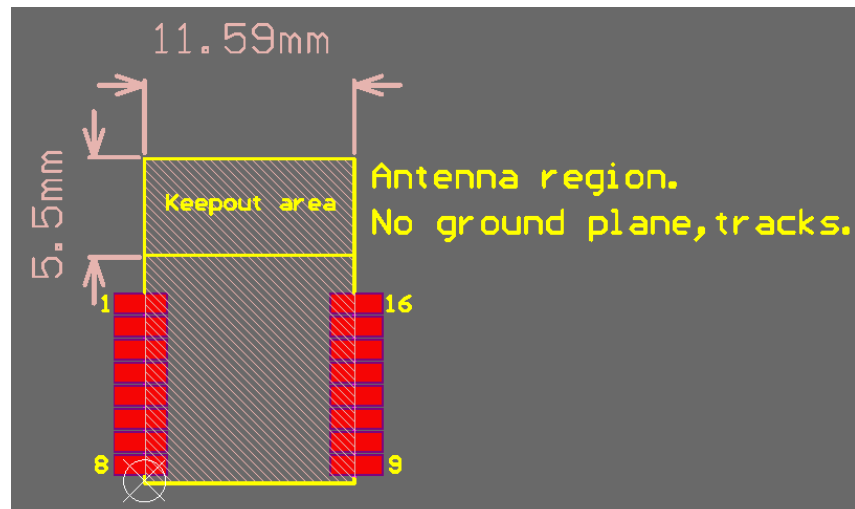
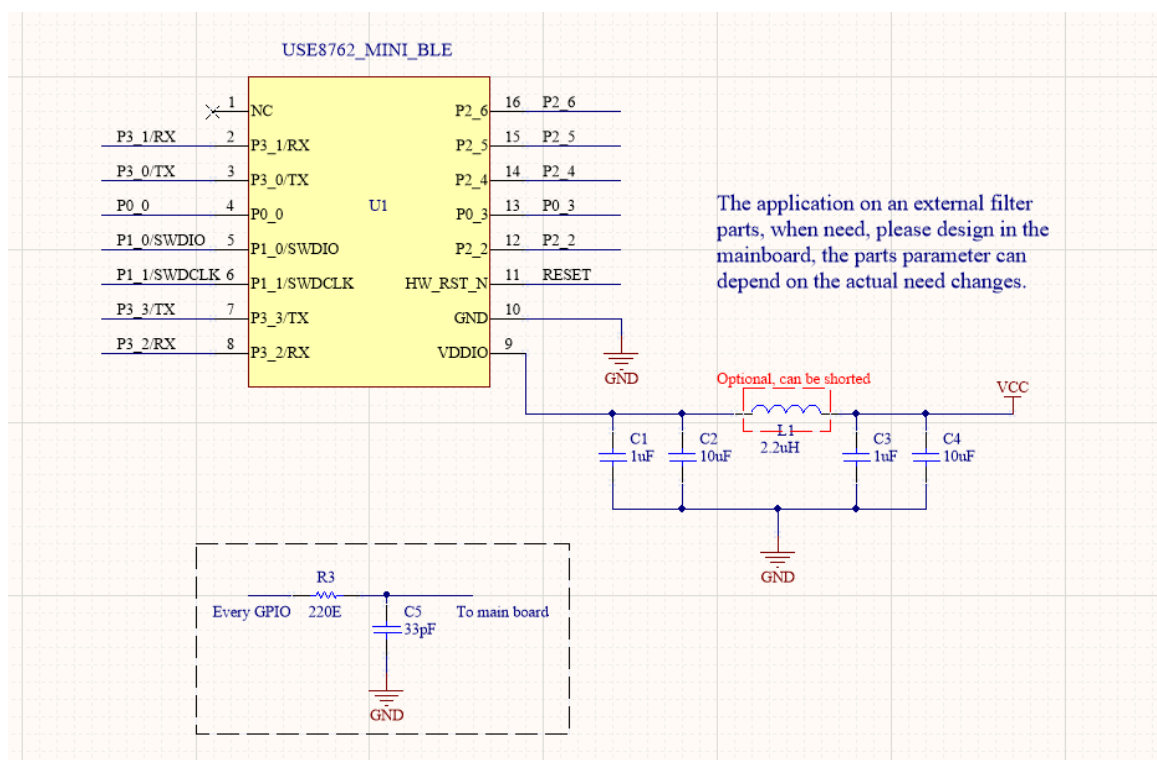


Fig 10 : Module Footprint Clearance

Recommended Connections:

The diagram (show in below Fig 10.) of the module application on external filter parts, when needed please design in the mainboard, the parts parameter can depend on the actual need to change. And pay attention to the increase and decrease ramping of supply voltage faster than 0.5v/ms;



Recommended Assembly:

Assembly recommendation

For smaller coin cell batteries, with high worst-case end-of-life equivalent source resistance, A 22- μ F VDDS input capacitor must be used to ensure compliance with this slew rate. Recommended one is to use MLCC for low ESR.

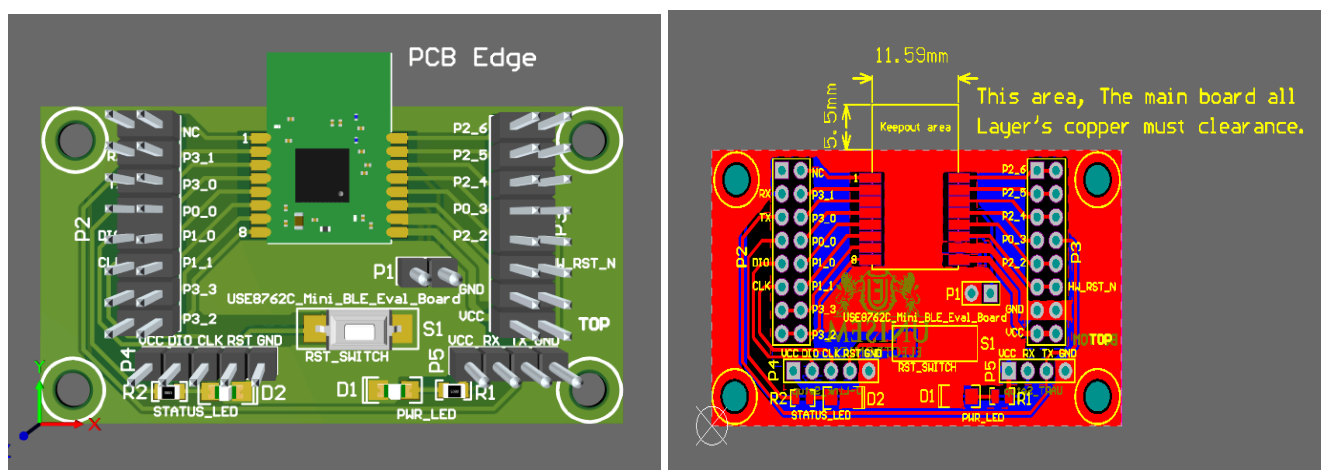


Fig 12 : Module Assembly

The module assembled in the motherboard, must pay attention to: Around the antenna and all layer's copper must be clear, the module antenna should be on the edge, and there is no metal or obstruct electromagnetic radiation material to surround the antenna area.

Module placement strategy:

BLE/Sub-1GHz modules are RF devices that require proper placement on PCB to ensure optimal performance. The antenna on the PCB has (about) an omnidirectional radiation pattern. To maximize antenna efficiency, an adequate grounding plane must be provided under the module. However, the areas underneath and surrounding the antenna area must be free of copper. The position of the module on the host board and overall design of the product enclosure contribute to antenna

performance. Poor design affects radiation patterns and can result in reflection, diffraction, and/or scattering of the transmitted signal thus limiting the range.

Basic guidelines:

- i. Never place the ground plane or route copper traces directly underneath the antenna portion of the module.
- ii. Never place the antenna close to metallic objects.
- iii. Keep wiring, components and objects away from the antenna.
- iv. Do not place the antenna in a metallic or metalized plastic enclosure.
- v. Enclosure walls should be 1cm or more away from the antenna in all directions.
- vi. If possible, mount the antenna overhanging the edge of the host board (Motherboard). Add an uninterrupted ground plane on the host board, directly underneath the top shielded zone of the module, up to the PCB edge. Adding a ground plane will allow traces to be run on the bottom side of the host board if required.
- vii. If the antenna cannot be mounted in an overhanging position, then provisions must be made to keep the area clear of copper.
- viii. If any components containing metal conductor or conductive substance are placed close to the antenna, it might obstruct radio wave radiation, which causes a significant reduction in communication distance significantly.

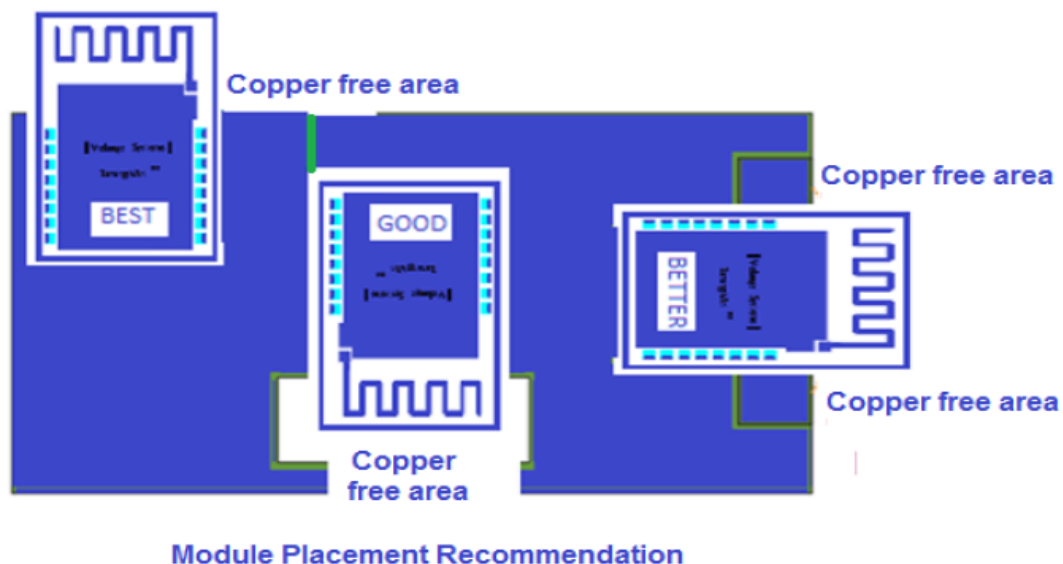


Fig 13

Recommended Reflow Profile:

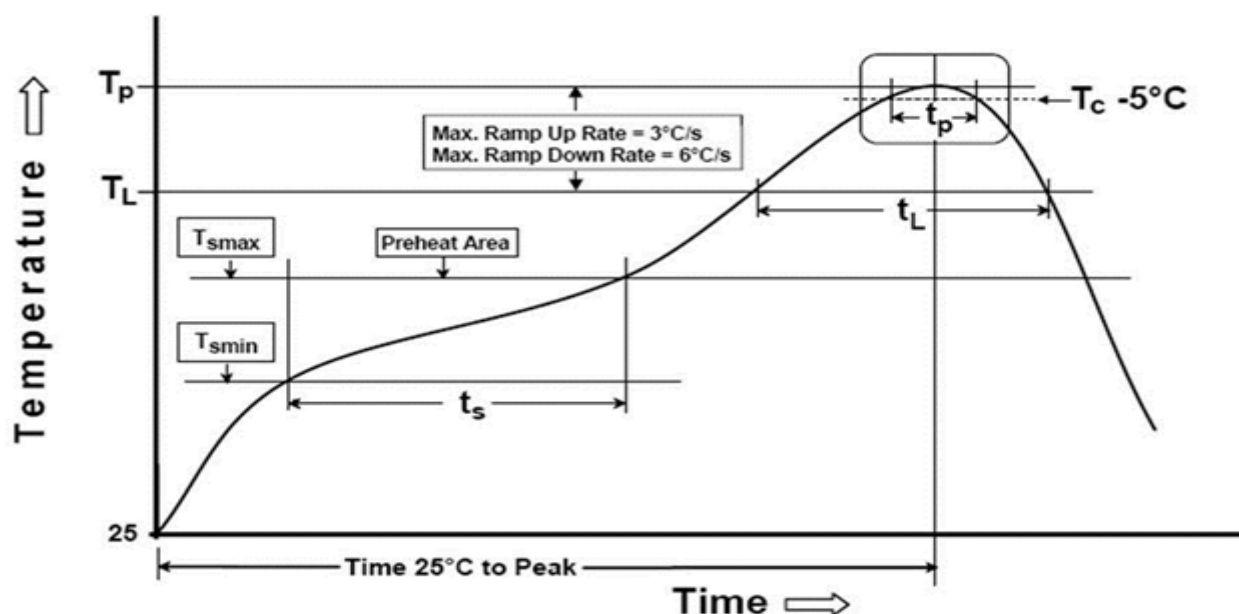


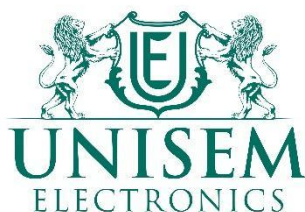
Fig 14 : Reflow Profile Plot

Stage	Note	Pb-Free Assembly
Average ramp-up rate	TL to Tp	3°C/ second max.
Preheat	Temperature min (T _{smin})	150 °C
	Temperature max (T _{smax})	200 °C
	Time (t _{smin} to t _{smax})	60 – 120 seconds
Time maintained above	Temperature (T _L)	217 °C
	Time (t _L)	60 – 150 seconds
Peak package body temperature (T _p)		See table below. Tp must not exceed the specified classification temperature in the table below.
Time(tp) within 5 °C of the specified classification temperature (Tc)		30 seconds

Ramp-down rate (TP to TL)	6 °C / seconds max.
Time 25 °C to peak temperature	8 minutes max.

Package Thickness	Volume < 350 mm ³	Volume 350 – 2000 mm ³	Volume > 2000 mm ³
< 1.6 mm	260 +0 /-5 °C	260 +0/-5 °C	260 +0 /-5 °C
1.6 – 2.5 mm	260 +0 /-5 °C	250 +0/-5 °C	245 +0/-5 °C
≥ 2.5 mm	250 +0 /-5 °C	245 +0/-5 °C	245 +0/-5 °C

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